

Table of contents

iMotions Lab > Analysis > Eye Tracking Analysis > Areas of Interest (AOIs)

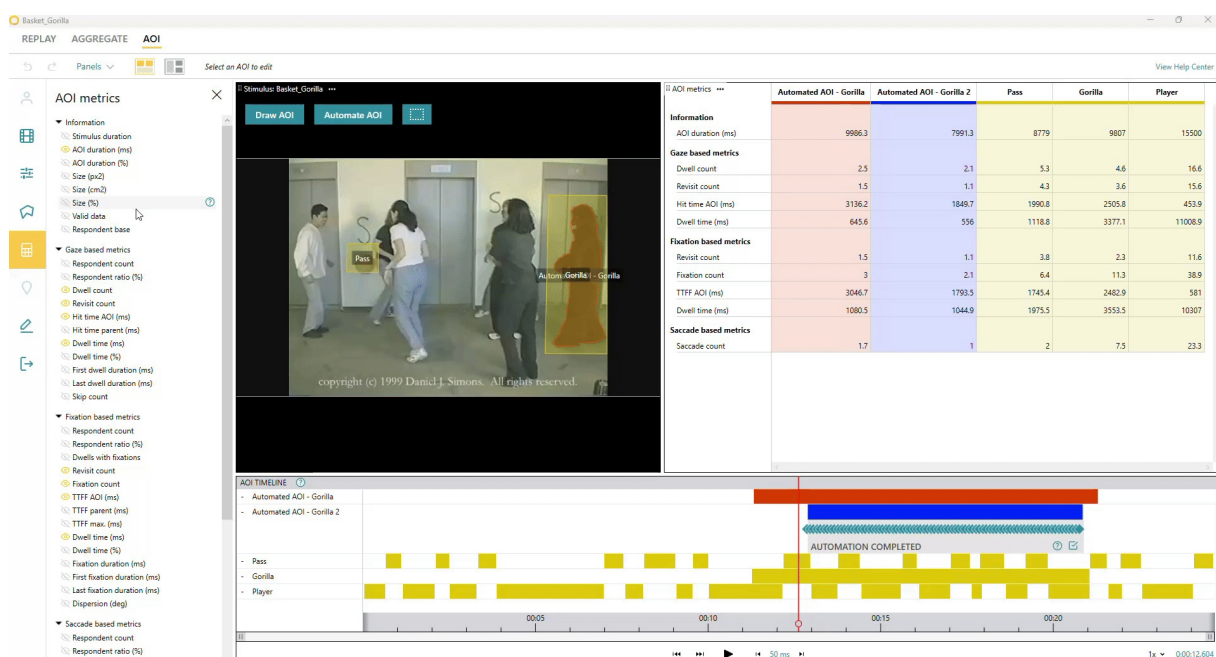
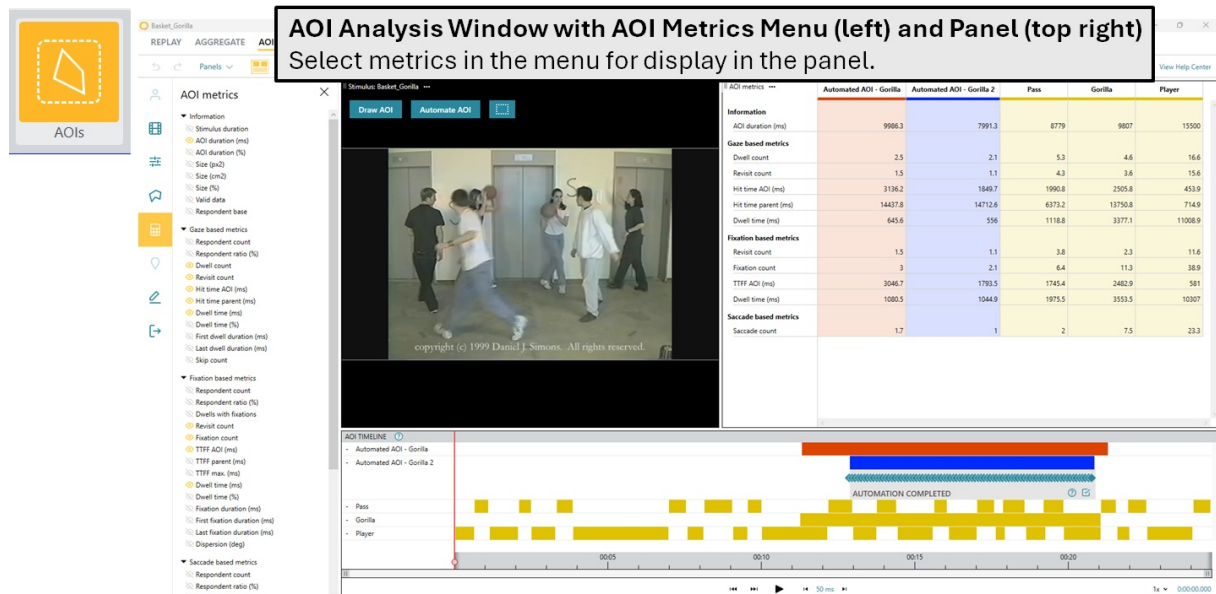
AOI Metrics 2

AOI Metrics

This article provides information about the eye-tracking metrics from Areas of Interest (AOIs) available in iMotions. [How to Create AOIs](#).

AOI Metrics are found in the **AOI Analysis Windows** and in [AOI Metrics Exports](#).

- Descriptions of each AOI metrics (48 total) can be found within the AOI Analysis window in iMotions - by hovering over the metrics and '?' icon.
- AOI Metrics are split into the following categories in the AOI Analysis Menu (drop-down lists on left), which controls displayed metrics in the AOI Metrics Panel (on the top-right).
 - **Information** - Details about the AOIs used to calculate metrics (based on AOI definitions).
 - **Gaze based metrics** - Metrics from gaze data (raw data without fixation classification).
 - **Fixation based metrics** - Metrics from fixation data (with fixation classification applied).
 - **Saccade based metrics** - Metrics calculated for saccades (via fixation classification).
 - **Mouse based metrics** - Metrics calculated from mouse clicks (based on respondent data/input).
- Note: 'NA' will appear if an AOI metric can not be calculated (e.g. when an AOI does not contain gaze/fixation data).
- [Detailed AOI Help Center](#) with information about related to AOI metrics and fixation classifiers.



Information

- **Stimulus duration** - The average duration of how long the stimulus was presented.
- **AOI duration (ms)** - The average duration of how long the AOI was active on the stimulus.
- **AOI duration (%)** - Average percentage of the stimulus duration during which the AOI was active.
- **Size (px2)** - Size of the AOI in square pixels.
- **Size (cm2)** - Size of the AOI in square centimeters.
- **Size (%)** - Percentage of the stimulus area covered by the AOI.
- **Valid data** - Average percentage of collected and interpolated samples. Values below 100% indicate missing data.

- **Respondent data** - Number of respondents for whom the AOI was defined.

Gaze-based metrics

- **Respondent count** - Number of respondents who gazed at the AOI.
- **Respondent ratio (%)** - Percentage of respondents who gazed at the AOI.
- **Dwell count** - Average of how often the gaze of the respondents entered the AOI (i.e. how often they visited the AOI on average).
- **Revisit count** - Average of how often the respondents looked back at the AOI after the first dwell.
- **Hit time AOI (ms)** - Average time that passed since the AOI's start time until the gaze entered the AOI for the first time.
- **Hit time parent (ms)** - Average time that passed since the start of the AOI's parent (the current Stimulus, Scene, or Gaze Map) until the gaze entered the AOI for the first time.
- **Dwell time (ms)** - Average of how long the respondents gazed at the AOI (i.e. Average of the total time gazed on the AOI across respondents).
- **Dwell time (%)** - Average of how long the respondents gazed at the AOI about the time during which the AOI was active.
- **First dwell duration (ms)** - Average duration of the respondents' very first dwell on the AOI.
- **Last dwell duration (ms)** - Average duration of the respondents' very last dwell on the AOI.
- **Skip count** - Average of how often the respondent gazed at the AOI without a detected fixation.

Fixation-based metrics

- **Respondent count** - Number of respondents who fixated on the AOI.
- **Respondent ratio (%)** - Percentage of respondents who fixated on the AOI.
- **Dwells with fixations** - Average of how often the respondents looked at the AOI and fixated on it at least once.
- **Revisit count** - Average of how often the respondents looked back and fixated on the AOI after the first dwell.
- **Fixation count** - Average amount of fixations detected inside the AOI.

- **TTFF AOI (ms)** - Average time that passed from the AOI's start time until the first fixation was detected on the AOI.
- **TTFF parent (ms)** - Average time that passed since the start of the AOI's parent (the current Stimulus, Scene, or Gaze Map) until the first fixation was detected on the AOI.
- **TTFF max. (ms)** - Average time that passed since the start of the AOI's parent (the current Stimulus, Scene, or Gaze Map) until the first fixation was detected on the AOI. In case the respondent did not fixate on the AOI, the parent's presentation duration (as maximally possible time) is used as TTFF (time to first fixation).
- **Dwell time (ms)** - Average of how long the respondents fixated at the AOI (i.e. Average of the total time fixated on the AOI across respondents).
- **Dwell time (%)** - Average of how long the respondents fixated on the AOI about the time during which the AOI was active.
- **Fixation duration (ms)** - Average duration of all fixations detected inside the AOI.
- **First fixation duration (ms)** - Average duration of the respondents' very first fixation inside the AOI.
- **Last fixation duration (ms)** - Average duration of the respondents' very last fixation inside the AOI.
- **Dispersion (deg)** - Average of how wide the gaze points of the respondents' fixations were spread out.

Saccade-based metrics

- **Respondent count** - Number of respondents who had a saccade within the AOI.
- **Respondent ratio (%)** - Percentage of respondents who had a saccade within the AOI.
- **Dwells with saccades** - Average of how often the respondents looked at the AOI with at least one saccade between entry and exit.
- **Saccade count** - Average amount of saccades detected inside the AOI.
- **Entry saccade onset AOI (ms)** - Average time that passed since the AOI's start time until the first saccade was directed towards the AOI.
- **Entry saccade onset parent AOI (ms)** - Average time that passed since the start of the AOI's parent (the current Stimulus, Scene, or Gaze Map) until the first saccade was directed towards the AOI.

- **Saccade duration (ms)** - Average duration of all saccades detected inside the AOI.
- **Amplitude (deg)** - Average amplitude of all saccades detected inside the AOI (i.e. angular distance that the eyes traveled from start point to end point).
- **Peak velocity (deg/s)** - Average peak velocity of all saccades detected inside the AOI (i.e. the maximal speed of the eyes during this saccade).
- **Peak acceleration (deg/s²)** - Average peak acceleration of all saccades detected inside the AOI (i.e. the maximal increase in speed of the eyes during this saccade).
- **Peak deceleration (deg/s²)** - Average peak deceleration of all saccades detected inside the AOI (i.e. the maximal decrease in speed of the eyes during this saccade).
- **Direction (deg)** - Average direction of all saccades detected inside the AOI, indicated as clockwise angles (0 degrees means a horizontal saccade from left to right, 90 degrees a vertical saccade from top to bottom).

Mouse-based metrics

- **Respondent count** - Number of respondents who clicked on the AOI.
- **Respondent ratio (%)** - Percentage of respondents who clicked on the AOI.
- **Mouse click count** - Average of all mouse clicks detected inside the AOI.